

BANJA LUKA, Bosnia and Herzegovina

WMO: 145420

Lat: **44.7938N**Lon: **17.2058E**Elev: **156**StdP: **99.46**Time Zone: **1.00 (EUC)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-7.6	-12.2	1.3	-9.3	-9.6	1.7	-5.9	8.5	12.9	6.7	10.1	1.0	290	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	35.7	21.9	33.7	21.7	31.9	21.3	23.6	32.1	22.6	31.3	21.8	30.3	2.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.8	27.9	19.8	14.8	26.7	19.0	14.0	25.4	71.0	32.2	67.3	31.4	64.2	30.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.0	5.3	4.1	DB	-15.2	38.2	4.5	2.3	-18.4	39.9	-21.0	41.2	-23.5	42.5	-26.7	44.1	
			WB	-15.4	24.9	4.3	0.8	-18.4	25.5	-20.9	26.0	-23.3	26.5	-26.4	27.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	1.8	2.8	7.6	13.1	16.9	21.2	23.1	22.8	17.7	12.6	8.2	2.8
	DBStd	8.67	5.09	5.20	5.03	3.46	3.81	3.86	3.32	3.43	3.92	4.06	4.68	4.50
	HDD10.0	916	258	204	106	13	2	0	0	0	0	21	86	227
	HDD18.3	2562	513	434	334	159	72	18	3	5	58	181	303	481
	CDD10.0	1868	3	3	30	105	216	336	407	397	230	102	33	5
	CDD18.3	473	0	0	0	2	28	104	152	144	38	3	1	0
	CDH23.3	5408	0	0	5	72	366	1151	1762	1576	421	54	2	0
	CDH26.7	2429	0	0	0	12	98	489	871	799	156	4	0	0
Wind	WSAvg	1.4	1.3	1.5	1.9	1.6	1.6	1.4	1.5	1.5	1.3	1.3	1.3	1.3
Precipitation	PrecAvg	1056	71	68	69	91	99	110	105	96	87	72	99	89
	PrecMax	1281	134	195	153	147	192	277	223	304	206	276	196	190
	PrecMin	685	9	14	16	26	14	59	36	6	19	0	26	8
	PrecStd	146	33	40	36	30	49	56	49	64	51	57	42	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.3	24.0	28.6	31.5	35.1	38.1	39.0	34.6	27.3	23.1	18.8
		MCWB	10.8	11.1	13.0	16.3	20.0	22.9	22.5	21.7	20.4	17.5	14.7	12.7
	2%	DB	14.9	16.1	21.6	25.3	29.3	33.3	35.7	36.1	30.6	25.1	21.0	16.3
		MCWB	9.6	9.7	11.8	14.7	19.5	22.2	22.0	21.5	19.3	16.8	13.9	10.7
	5%	DB	12.3	13.8	19.3	23.2	27.3	31.5	33.6	33.7	27.8	22.8	18.2	13.4
		MCWB	8.1	8.5	11.1	14.0	18.3	21.5	22.0	21.2	18.6	16.1	12.5	9.0
	10%	DB	9.2	11.1	16.9	20.9	25.2	29.6	31.6	31.4	25.4	20.3	15.6	9.4
		MCWB	6.3	7.3	10.1	13.0	17.3	20.7	21.5	20.7	18.0	15.1	11.7	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	11.6	13.7	17.3	21.6	24.6	24.9	24.4	21.6	18.8	15.6	13.1
		MCDB	16.8	18.5	21.8	26.6	29.8	32.8	32.9	32.7	30.0	25.2	21.0	18.2
	2%	WB	9.8	10.0	12.5	15.6	20.3	23.1	23.7	23.0	20.3	17.4	14.3	11.0
		MCDB	14.3	15.2	19.8	23.1	27.8	31.1	32.2	32.2	28.3	23.2	19.2	15.7
	5%	WB	8.4	8.8	11.5	14.6	19.0	22.1	22.8	22.1	19.4	16.4	13.3	9.1
		MCDB	12.0	13.1	18.0	21.0	25.6	30.2	31.4	31.5	26.2	21.6	17.9	12.7
	10%	WB	6.6	7.5	10.4	13.7	17.8	21.2	21.9	21.2	18.4	15.4	11.9	7.2
		MCDB	9.0	10.8	16.3	19.6	24.0	28.4	30.4	29.7	23.9	19.9	15.4	9.3
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	11.0	12.2	11.9	12.0	13.0	13.1	11.5	10.8	9.0	7.1
		MCDBR	13.0	14.4	17.1	17.2	15.9	15.7	17.1	18.3	16.4	15.2	14.3	13.2
		MCWBR	8.2	8.6	8.8	8.1	7.3	6.4	6.1	6.4	6.8	7.9	8.2	8.2
	5% WB	MCDBR	12.2	13.1	14.9	15.1	14.5	14.6	15.1	15.8	14.1	14.1	12.5	12.1
		MCWBR	7.8	8.0	8.0	7.5	7.1	6.3	6.2	6.3	6.2	7.7	7.4	7.7
Clear-Sky Solar Irradiance	taub	0.328	0.357	0.392	0.443	0.433	0.432	0.432	0.432	0.416	0.386	0.349	0.329	
	taud	2.398	2.311	2.254	2.132	2.212	2.257	2.254	2.260	2.279	2.360	2.416	2.414	
	Ebn at Noon	778	816	834	818	840	841	835	820	800	773	745	737	
	Edh at Noon	76	98	119	146	139	134	132	127	115	93	74	68	
All-Sky Solar Radiation	RadAvg	1.28	1.95	3.13	4.24	5.26	5.98	6.11	5.44	3.72	2.53	1.49	1.07	
	RadStd	0.18	0.28	0.44	0.53	0.47	0.50	0.43	0.61	0.46	0.30	0.18	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (30 neighbors)	+0.50	N/A	N/A	N/A	+0.35	N/A	N/A	-120	N/A	N/A

Nomenclature: See separate page